



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2014
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #2

Mathematics Extension 2

General Instructions

- Reading Time – 5 Minutes
- Working time – 2 Hours
- Write using black or blue pen. Pencil may be used for diagrams.
- Each of the three Sections (A, B, and C) is to be returned in a separate bundle.
- All necessary working should be shown in every question.

Total Marks – 91

- Attempt questions 1 – 6
- Board approved calculators maybe used.
- Full marks may not be awarded for careless or badly arranged work.
- Unless otherwise stated, give answers in simplest exact form.

Examiner: *P.R.Bigelow*

Section A
(Start a new answer sheet.)

Question 1. (17 marks)

- | | Marks |
|--|--------------|
| (a) Find $\int \frac{2x+1}{x^2+2x+2} dx$. | 2 |
| (b) Evaluate $\arg((2+i)\bar{z})$ where $z = -1-3i$ | 2 |
| (c) (i) Sketch the curve:
$(x+1)(y-2)=1$ | 2 |
| (ii) Express the function in (i) in the form $y=f(x)$.
Hence or otherwise sketch: | |
| (α) $y=f(x-2)$ | 1 |
| (β) $y^2=f(x)$ | 2 |
| (γ) $y=\frac{1}{f(x)}$ | 2 |
| (d) Find:
$\int \frac{d\theta}{8\cos^2\theta+1}$ | 2 |
| (e) If z is a complex number such that $z=k(\cos\theta+i\sin\theta)$ where k is real, show that
$\arg(z+k)=\frac{\theta}{2}.$ | 2 |
| (f) Use the substitution $u=1-\sqrt{x}$ to find:
$\int \frac{dx}{1-\sqrt{x}}$ | 2 |

Question 2. (15 marks)

Marks

- (a) (i) Show that:

2

$$\int_{-a}^a f(x) dx = \int_0^a [f(x) + f(-x)] dx$$

- (ii) Hence evaluate

$$\int_{-2}^2 \frac{dx}{1+e^{-x}}$$

2

- (b) (i) Express $-\sqrt{27} - 3i$ in mod-arg form.

2

- (ii) Hence find $(-\sqrt{27} - 3i)^6$, giving your answer in the form $a + ib$ where a and b are real.

2

- (c) The complex number z lies on the locus $\arg(z+i) = \frac{\pi}{4}$:

- (i) Sketch the locus, showing any intercepts with the axes.

1

- (ii) Find the least value of $|z|$.

2

- (d) (i) Use the graph of $y = \sin x$ to sketch on separate diagrams, the graphs of the functions $y = |\sin x|$ and $y = \sin|x|$ for $-2\pi \leq x \leq 2\pi$.

2

- (ii) Evaluate:

$$\int_{-2\pi}^{2\pi} (\sin|x| + |\sin x|) dx$$

2

Section B
(Start a new answer sheet.)

Question 3. (16 marks)

	Marks
(a) The nine letters of the word <i>REDIVIDER</i> are arranged at random in a line.	
(i) How many letter sequences are possible?	2
(ii) What is the probability that the sequence is the same from left to right as from right to left.	2
(b) An object of mass 10 kg falls from rest from a stationary balloon, and experiences air resistance of magnitude $4v^2$. Take the value of g to be 10 m/s^2 , and take downwards to be positive.	2
(i) Show that the equation of motion is:	1
$\ddot{x} = \frac{50 - 2v^2}{5}$	
(ii) Find the terminal velocity of the object.	1
(iii) Show that the velocity v at time t is given by:	3
$t = \frac{1}{4} \ln \left(\frac{5+v}{5-v} \right)$	
(iv) Show $v = \frac{5(e^{4t} - 1)}{e^{4t} + 1}$	2
(v) Find the vertical distance fallen as a function of t .	3

Question 4 (14 marks)

(a) Let $p = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$

(i) Show that $1 + p + p^2 + p^3 + p^4 = 0$ 2

(ii) Simplify $(p + p^4)(p^2 + p^3)$ 1

(iii) Form a quadratic equation with roots $(p + p^4)$ and $(p^2 + p^3)$ 1

(iv) Hence or otherwise show that $\cos \frac{2\pi}{5} \cdot \cos \frac{4\pi}{5} = -\frac{1}{4}$ 2

(b) There are eight people, 5 men and 3 women, to be seated at a round table. How many arrangements are there if:

(i) there are no restrictions? 1

(ii) no two women are next to one another? 2

(Give your answers in the form of powers and factorials.)

(c) The constants m and n are such that the equation $x^3 - mx + n = 0$ has three no-zero roots α, β, γ .

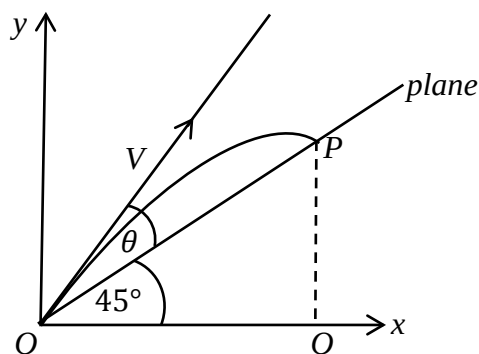
(i) Find in terms of m and n the cubic equation (in expanded form) whose roots are α^3, β^3 and γ^3 . 2

(ii) Hence or otherwise find $\alpha^7\beta\gamma + \alpha\beta^7\gamma + \alpha\beta\gamma^7$ in terms of m and n . 3

Section C
(Start a new answer booklet)

Question 5 (14 marks)

(a)



A particle is projected from a point O on a plane inclined at 45° to the horizontal. The velocity of projection is V at an angle of θ to the inclined plane, where $\theta = \tan^{-1} \frac{1}{2}$.

[You are given that $y = x \tan \alpha - \frac{gx^2}{2V^2} (1 + \tan^2 \alpha)$ where $\alpha = 45^\circ + \theta$]

(i) Show that $\tan \alpha = 3$. 1

(ii) Hence show that $y = 3x - \frac{5gx^2}{V^2}$. 1

(iii) Show that the range OP is $\frac{2\sqrt{2}V^2}{5g}$. 2

(iv) Show that it meets the plane at right-angles. 3

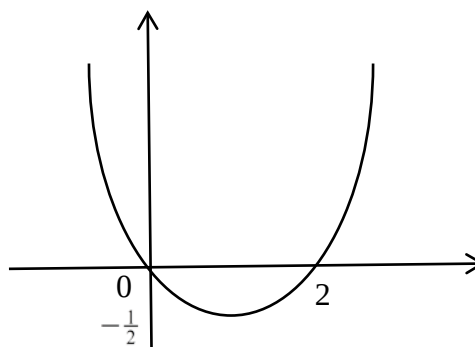
(b) The sketch is of the parabola

$$f(x) = \frac{x^2 - 2x}{2}.$$

Without calculus, draw the following on separate diagrams, showing essential features:

(i) $y = |f(x)|$

(ii) $y = \frac{x}{2} |x - 2|$



(iii) $y = \tan^{-1}(f(x))$

(iv) $|y| = f(x)$

Question 6 (15 marks)

(a) Sketch the following, showing essential features:

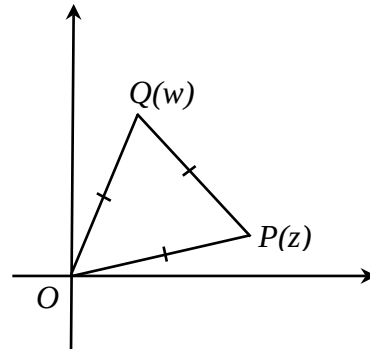
(i) $y = \ln(\ln x)$ 2

(ii) $f(x) = \cos^{-1}(\sin x)$ 2

(b) The triangle OPQ is equilateral.

P represents the complex number z , and Q represents the complex number w .

Show that $z^3 + w^3 = 0$.



(c) Let $I_n = \int_0^1 (1-x^2)^n dx$ and $J_n = \int_0^1 x^2 (1-x^2)^n dx$.

(i) Using integration by parts, on I_n , show that $I_n = 2nJ_{n-1}$. 2

(ii) Hence show that $I_n = \frac{2n}{2n+1} I_{n-1}$. 2

(iii) Show that $J_n = I_n - I_{n+1}$. 2

(iv) Hence deduce that $J_n = \frac{1}{2n+3} I_n$. 2

(v) Hence find a reduction formula for J_n in terms of J_{n-1} . 1

This is the end of the paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

SBHS 2014 Extension 2 Assessment #2 Solutions

1) a) $\int \frac{2x+1}{x^2+2x+2} dx$

$$= \int \left(\frac{2x+2}{x^2+2x+2} - \frac{1}{x^2+2x+1+1} \right) dx$$

$$= \int \left(\frac{2x+2}{x^2+2x+2} - \frac{1}{(x+1)^2+1} \right) dx$$

$$= \ln(x^2+2x+2) - \tan^{-1}(x+1) + C$$

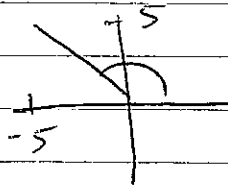
b) $\arg((2+i)(-1-3i))$

$$= \arg((2+i)(-1+3i))$$

$$= \arg(-2+6i-i-3)$$

$$= \arg(-5+5i)$$

$$= \frac{3\pi}{4}$$



c) i) $(x+1)(y-2) = 1$

$$x+1 \neq 0$$

$$x \neq -1$$

$$y-2 \neq 0$$

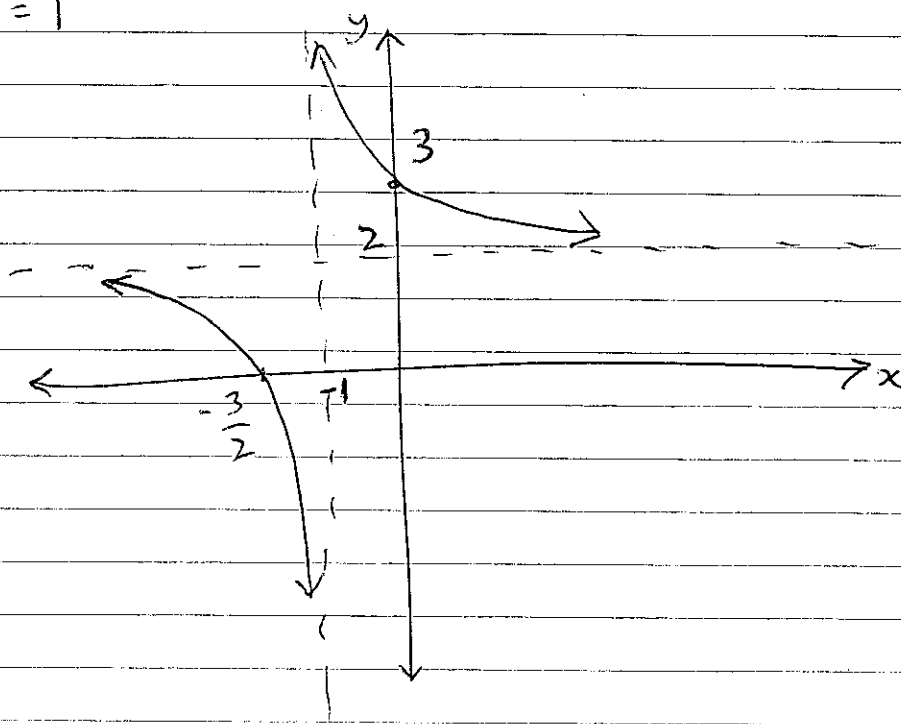
$$y \neq 2$$

$$\text{when } x=0$$

$$y=3$$

$$\text{when } y=0$$

$$x = -\frac{3}{2}$$

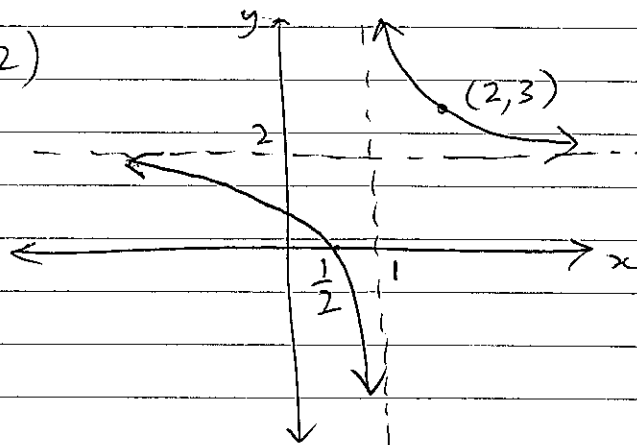


$$\text{ii) } (x+1)(y-2) = 1$$

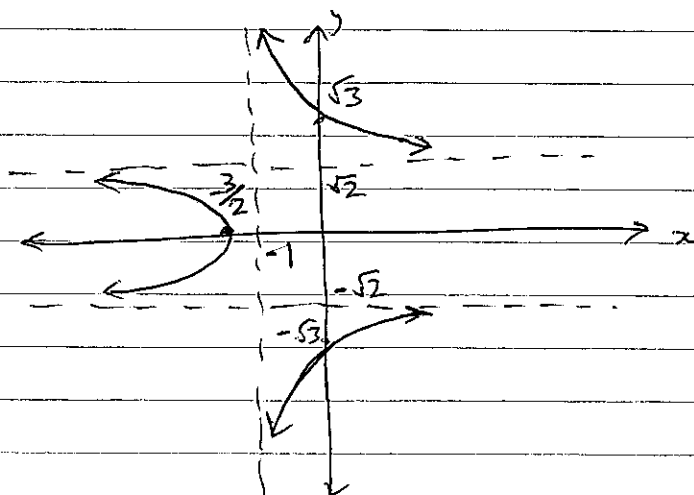
$$y-2 = \frac{1}{x+1}$$

$$y = \frac{1}{x+1} + 2$$

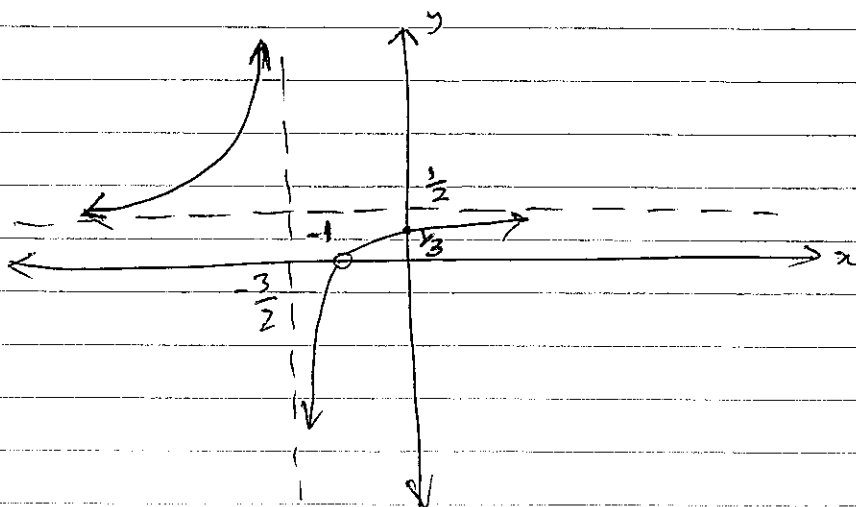
$$\alpha) y = f(x-2)$$



$\beta)$



$\gamma)$



$$d) \int \frac{d\theta}{8\cos^2\theta + 1}$$

$$= \int \frac{d\theta}{8\cos^2\theta + \sin^2\theta + \cos^2\theta}$$

$$= \int \frac{d\theta}{9\cos^2\theta + \sin^2\theta}$$

$$= \int \frac{\frac{1}{\cos^2\theta}}{\frac{9\cos^2\theta}{\cos^2\theta} + \frac{\sin^2\theta}{\cos^2\theta}} d\theta$$

$$= \int \frac{\sec^2\theta d\theta}{9 + \tan^2\theta}$$

$$\text{let } u = \tan\theta$$

$$\frac{du}{d\theta} = \sec^2\theta$$

$$d\theta = \frac{du}{\sec^2\theta}$$

$$= \int \frac{\cancel{\sec^2\theta}}{9 + u^2} \cdot \frac{du}{\cancel{\sec^2\theta}} \quad a=3$$

$$= \frac{1}{3} \tan^{-1}\left(\frac{u}{3}\right) + C$$

$$= \frac{1}{3} \tan^{-1}\left(\frac{\tan\theta}{3}\right) + C$$

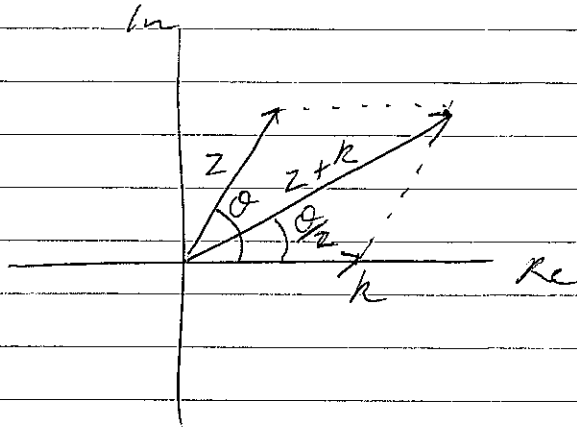
$$\begin{aligned} e) \quad \arg(z+k) &= \arg(k(\cos\theta + i\sin\theta) + k) \\ &= \arg(k(\cos\theta + 1) + ki\sin\theta) \\ &= \arg(k(2\cos^2\frac{\theta}{2} - 1 + 1) + ki \cdot 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}) \\ &= \arg(2k\cos\frac{\theta}{2}(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2})) \\ &= \frac{\theta}{2} \end{aligned}$$

Graphically

Note: $|z| = k$

$$|k| = k$$

$\therefore z+k$ is the diagonal of a rhombus.



since the diagonals of a rhombus bisect the angles, $\arg(z+k) = \frac{\theta}{2}$.

$$f) \int \frac{dx}{1-\sqrt{x}}$$

$$\text{let } u = 1 - \sqrt{x}$$

$$\frac{du}{dx} = -\frac{1}{2\sqrt{x}}$$

$$dx = -2\sqrt{x} du$$

$$= \int \frac{-2\sqrt{x} du}{u}$$

$$= \int \frac{-2(1-u)}{u} du$$

$$= 2 \int \frac{u-1}{u} du$$

$$= 2 \int \left(1 - \frac{1}{u}\right) du$$

$$= 2 [u - \ln u] + C$$

$$= 2 [1 - \sqrt{x} - \ln(1 - \sqrt{x})] + C$$

$$2) a) i) \int_{-a}^a f(x) dx$$

$$= \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$$

consider $\int_{-a}^0 f(x) dx$

let $u = -x$ when $x = -a, u =$

$\frac{du}{dx} = -1$ $x = 0, u = 0$

$dx = -du$

$$= \int_a^0 f(-u) \cdot -du$$

$$= - \int_a^0 f(-u) du$$

$$= \int_0^a f(-u) du$$

$$= \int_0^a f(-x) dx$$

for definite integrals
the variable can be changed

$$= \int_0^a f(-x) dx + \int_0^a f(x) dx$$

$$= \int_0^a [f(x) + f(-x)] dx$$

$$ii) \int_{-2}^2 \frac{dx}{1+e^{-x}}$$

$$= \int_0^2 \left[\frac{1}{1+e^{-x}} + \frac{1}{1+e^x} \right] dx$$

$$= \int_0^2 \left[\frac{e^x}{e^x+1} + \frac{1}{1+e^x} \right] dx$$

$$= \int_0^2 \left[\frac{1+e^x}{1+e^x} \right] dx$$

$$= \left[x \right]_0^2$$

$$= (2) - (0)$$

$$= 2$$

b) i) $-\sqrt{27} - 3i$

$$r^2 = (\sqrt{27})^2 + (3)^2$$

$$= 36$$

$$\underline{r = 6}$$

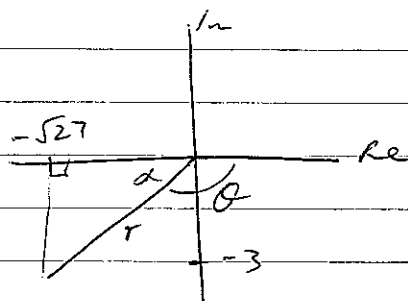
$$\tan \alpha = \frac{3}{\sqrt{27}}$$

$$= \frac{3}{3\sqrt{3}}$$

$$= \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\pi}{6}$$

$$\therefore \theta = -\frac{5\pi}{6}$$



$$-\sqrt{27} - 3i = 6 \left(\cos\left(-\frac{5\pi}{6}\right) + i \sin\left(-\frac{5\pi}{6}\right) \right)$$

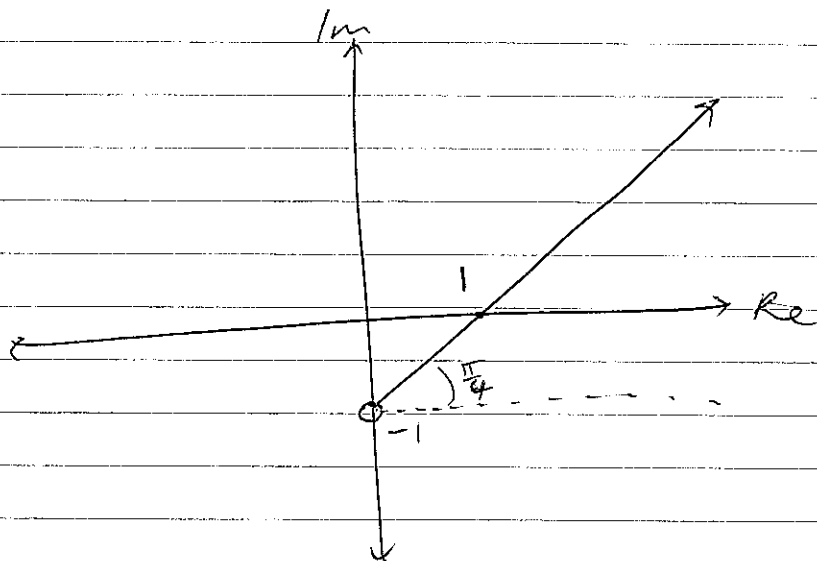
$$\text{ii) } (-\sqrt{27} - 3i)^6 = \left[6 \left(\cos\left(-\frac{5\pi}{6}\right) + i \sin\left(-\frac{5\pi}{6}\right) \right) \right]^6$$

$$= 6^6 \left(\cos(-5\pi) + i \sin(-5\pi) \right) \quad \text{using de Moivre's theorem}$$

$$= 46656((-1) + i(0))$$

$$= -46656$$

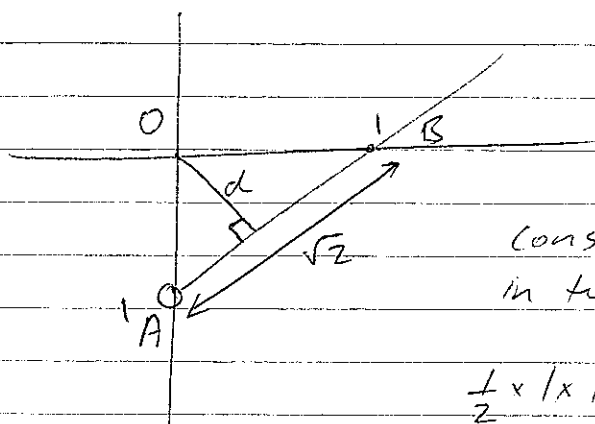
c) i)



$$\arg(z+i) = \frac{\pi}{4}$$

$$= \arg(z - (-i)) = \frac{\pi}{4}$$

ii)



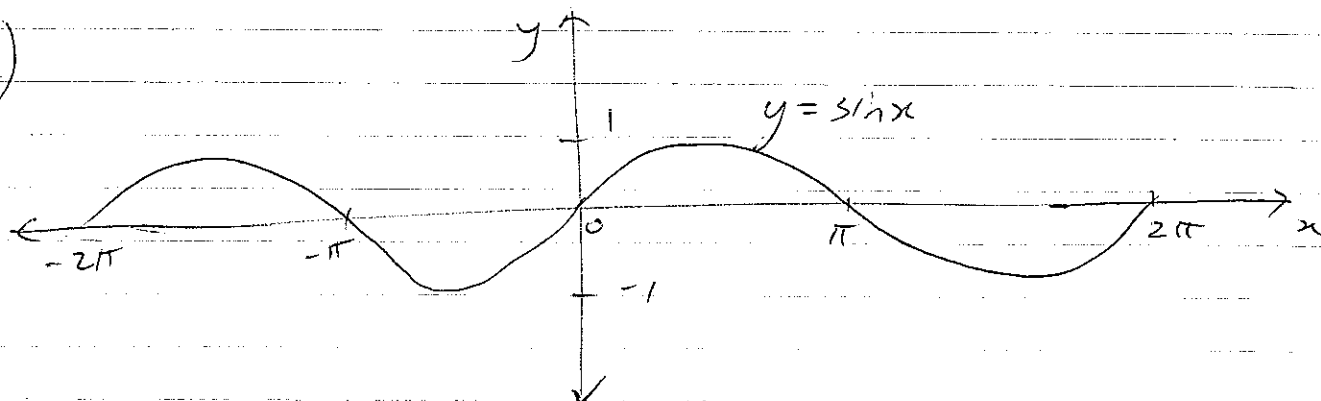
consider the area of $\triangle OAB$
in two ways

$$\frac{1}{2} \times |x| = \frac{1}{2} \times \sqrt{2} \times d$$

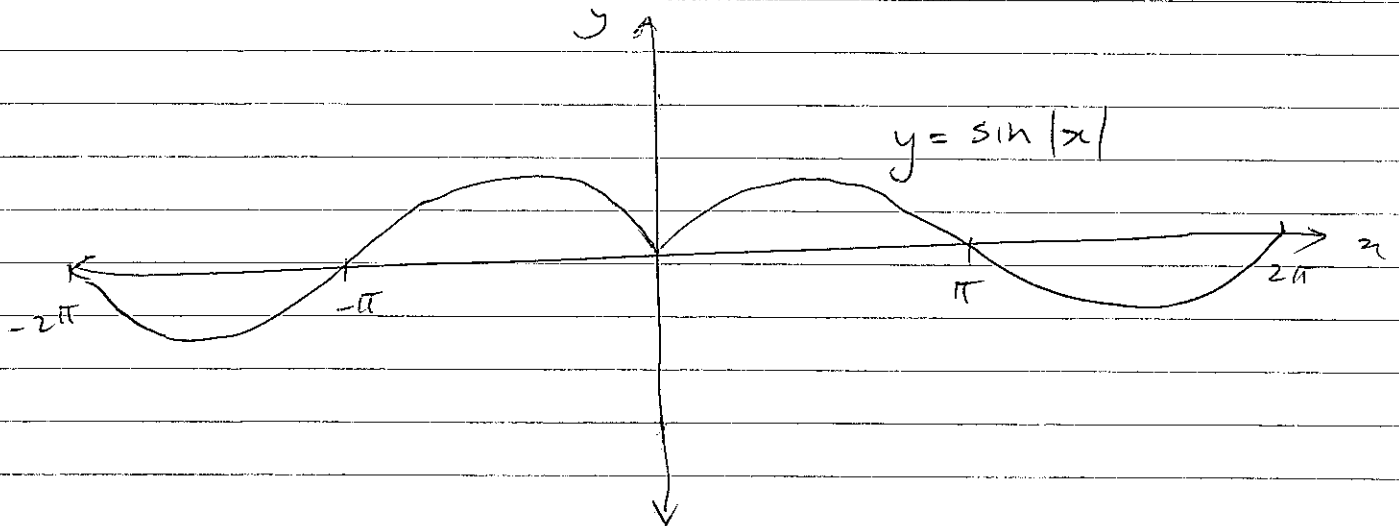
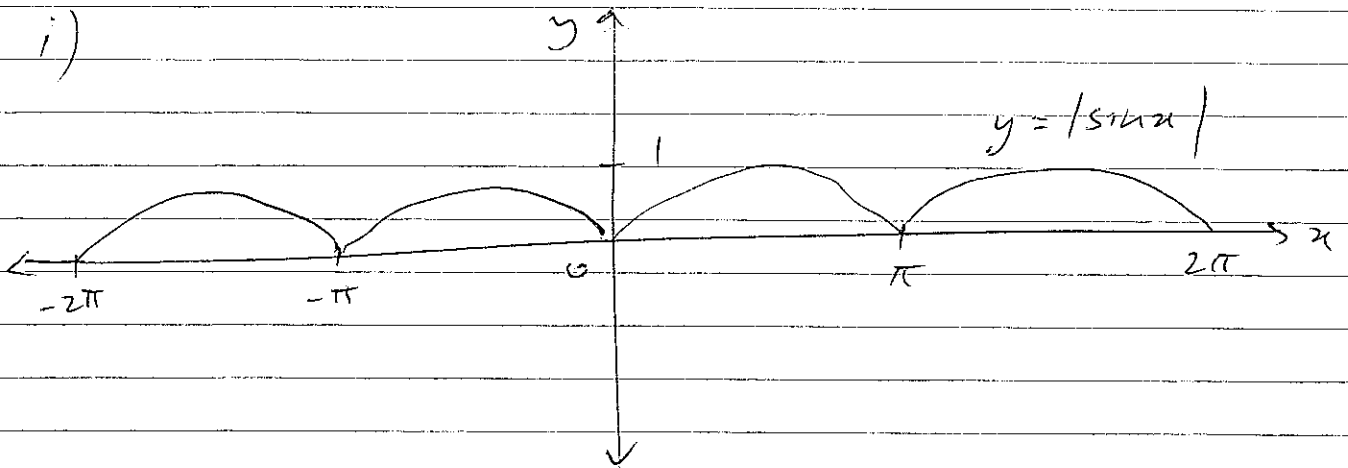
$$d = \frac{1}{\sqrt{2}}$$

$$\begin{aligned} \text{the least value of } |z| &= \frac{1}{\sqrt{2}} \\ &= \frac{\sqrt{2}}{2} \end{aligned}$$

d)



i)



ii)

$$\int_{-2\pi}^{2\pi} (\sin |x| + |\sin x|) dx$$

$$= \int_{-2\pi}^{2\pi} \sin |x| dx + \int_{-2\pi}^{2\pi} |\sin x| dx$$

$$= 0 + 4 \int_0^{\pi} \sin x dx$$

$$= 4 \left[-\cos x \right]_0^{\pi}$$

$$= 4 \left[-\cos \pi - (-\cos 0) \right]$$

$$= 4 \left[-(-1) + 1 \right]$$

$$= 8$$

Solutions

Question (3)

$V \rightarrow 1, D, E, R, I \rightarrow 2$

(i) $\therefore$ The possible sequences

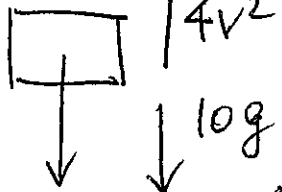
are: $\boxed{\frac{9!}{2^4} = \frac{362880}{16} = 22680}$

(ii) One way to put 'V' in the middle. The rest of D, E, R, I can be arranged 4! ways.

Now, once the letters on either the left/ right of V are determined there is only 1 way to arrange right/left side = $1 \times 4!$

$\therefore$ Prob. = $\boxed{\frac{4!}{(9!/2^4)} = \frac{1}{945}}$
 $= \boxed{0.0010582}$

(b) (i)



$$10\ddot{x} = 10g - 4v^2$$

$$\ddot{x} = g - \frac{2v^2}{5}$$

$$\boxed{\ddot{x} = \frac{50 - 2v^2}{5}} \quad g = 10$$

(ii) Terminal velocity $\boxed{\text{When } \ddot{x} = 0.}$
 $\Rightarrow \boxed{V = 5 \text{ m/sec.}}$

(iii) $\frac{dv}{dt} = \frac{50 - 2v^2}{5} \Rightarrow \frac{dt}{dv} = \frac{5}{50 - 2v}$
 $\therefore t = \frac{5}{2} \int \frac{dv}{25 - v^2}$

Now $\frac{1}{25 - v^2} = \frac{a}{5 + v} + \frac{b}{5 - v}$
 $\therefore 1 = a(5 - v) + b(5 + v)$

Equate coeff. of v $b - a = 0 \Rightarrow a = b$.
 Equate constant term $5a + 5b = 1$
 $\therefore a + b = \frac{1}{5}, 2a = \frac{1}{5}$
 $\therefore a = b = \frac{1}{10}$

$$\therefore t = \frac{5}{20} \int \left(\frac{1}{5+v} + \frac{1}{5-v} \right) dv$$

$$\Rightarrow t = \frac{1}{4} [\ln(5+v) - \ln(5-v)]$$

$$\therefore t = \frac{1}{4} \ln \left(\frac{5+v}{5-v} \right)$$

(iv) Now, $4t = \ln \left(\frac{5+v}{5-v} \right)$

$$\therefore \frac{5+v}{5-v} = e^{4t}$$

$$5e^{4t} - (e^{4t})v = 5 + v$$

$$5(e^{4t} - 1) = v(1 + e^{4t})$$

$$\therefore v = \frac{5(e^{4t} - 1)}{e^{4t} + 1}$$

(vi)

Now, $v = \frac{dx}{dt}$

$$\therefore \frac{dx}{dt} = \frac{5(e^{4t} - 1)}{e^{4t} + 1}$$

Separating the variables

$$\int dx = 5 \int \left[\frac{(e^{4t} + 1) - 2}{e^{4t} + 1} \right] dt$$

$$= 5 \int \left(1 - \frac{2}{e^{4t} + 1} \right) dt$$

$$= 5t - 10 \int \frac{dt}{e^{4t} + 1}$$

Now, $10 \int \frac{dt}{e^{4t} + 1} \left(\times \frac{e^{-4t}}{e^{-4t}} \right)$

$$= 10 \int \frac{e^{-4t}}{1 + e^{-4t}} dt$$

Let $u = 1 + e^{-4t}$ then $du = -4e^{-4t} dt$

$$= -\frac{5}{2} \int \frac{(-4e^{-4t}) dt}{1 + e^{-4t}}$$

$$= -\frac{5}{2} \int \frac{du}{u} = -\frac{5}{2} \ln u$$

$$= -\frac{5}{2} \ln(1 + e^{-4t})$$

$$\therefore x = 5t - \frac{5}{2} \ln(1 + e^{-4t}) + c$$

When $t=0$, $x=0$, $\Rightarrow c = \frac{5}{2} \ln 2$

$$\therefore x = 5t + \frac{5}{2} \ln \left(\frac{2}{1 + e^{-4t}} \right)$$

$$\doteq 5.17$$

• Method (1). $\varphi(3) (b) (v)$

$$v = \frac{5(e^{4t} - 1)}{e^{4t} + 1}$$

$$\frac{dx}{dt} = \frac{5(e^{4t} - 1)}{e^{4t} + 1}$$

$$x = 5 \int \left(\frac{e^{4t} + 1}{e^{4t} + 1} - 2 \right) dt$$

$$= 5 \int \left(1 - \frac{2}{e^{4t} + 1} \right) dt$$

$$= 5 \int \left(1 + \frac{-2e^{-4t}}{1 + e^{-4t}} \right) dt$$

$$= 5 \int \left(1 + \frac{1}{2} \left(\frac{-4e^{-4t}}{1 + e^{-4t}} \right) \right) dt$$

$$= 5t + \frac{5}{2} \ln(1 + e^{-4t}) + C$$

When $t=0$, $x=0$, $0 = \frac{5}{2} \ln 2 + C$

$$\therefore x = 5t + \frac{5}{2} \ln \left(\frac{1 + e^{-4t}}{2} \right)$$

• Method (2) $v = 5 \int \frac{e^{4t} - 1}{e^{4t} + 1} dt$ ①

but $\therefore x = 5 \int \frac{e^{2t} - e^{-2t}}{e^{2t} + e^{-2t}} dt$

Let $u = e^{2t} + e^{-2t}$, $du = 2(e^{2t} - e^{-2t}) dt$

$$\therefore x = \frac{5}{2} \int \frac{du}{u}$$

$$\therefore x = \frac{5}{2} \ln(e^{2t} + e^{-2t}) + C$$

When $t=0$, $x=0$

$$\Rightarrow C = -\frac{5}{2} \ln 2$$

$$\therefore x = \frac{5}{2} \ln \left(\frac{e^{2t} + e^{-2t}}{2} \right)$$
 ②

Method (3) Q(3) (b) (v)

$$x = 5 \int \frac{e^{4t}}{e^{4t}+1} dt - 5 \int \frac{dt}{e^{4t}+1}$$
$$= \frac{5}{4} \ln(e^{4t}+1) - 5 \int \frac{4e^{4t} dt}{4e^{4t}(e^{4t}+1)}$$

$$\boxed{-\frac{5}{4} \int \frac{4e^{4t} dt}{e^{4t}(e^{4t}+1)}}$$

Now, to integrate this,
let $u = e^{4t}$, $du = 4e^{4t} dt$.
by partial fractions

$$\frac{1}{u(u+1)} = \frac{1}{u} - \frac{1}{u+1}$$

$$\therefore -\frac{5}{4} \int \left(\frac{4e^{4t} dt}{e^{4t}+1} \right)$$
$$= -\frac{5}{4} (\ln u - \ln(u+1))$$

$$= -\frac{5}{4} \ln \left(\frac{e^{4t}}{e^{4t}+1} \right)$$

$$\therefore x = \frac{5}{4} \ln(e^{4t}+1) - \frac{5}{4} \ln \left(\frac{e^{4t}}{e^{4t}+1} \right) + C$$

$$x = \frac{5}{4} \left[\ln(e^{4t}+1) - \ln(e^{4t}) + \ln(e^{4t}+1) \right] + C.$$

$$x = \frac{5}{4} (2 \ln(e^{4t}+1) - 4t) + C$$

When $t = 0$, $x = 0$

$$0 = \frac{5}{4} (2 \ln 2) + C$$

$$\Rightarrow C = -\frac{5}{2} \ln 2.$$

$$\therefore \boxed{x = \frac{5}{2} \ln \left(\frac{e^{4t}+1}{2} \right) - 5t}$$

Conclusion: (All 3 are
Now, expanding SAME because
of the log
expansions)

(1) method (1) we have

$$x = 5t + \frac{5}{2} \ln \left(\frac{1 + e^{4t}}{2} \right)$$

$$\therefore x = 5t + \frac{5}{2} \ln \left(\frac{e^{4t} + 1}{2e^{4t}} \right)$$

(2) Expand method (2)

$$x = \frac{5}{2} \ln \left(\frac{e^{2t} + e^{-2t}}{2} \right)$$

$$x = \frac{5}{2} \ln \left(\frac{e^{4t} + 1}{2e^{2t}} \right)$$

(3) Method (3) left

as it is

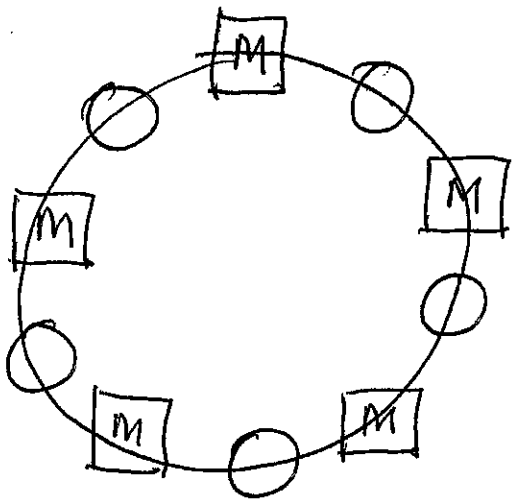
$$\therefore x = \frac{5}{2} \ln \left(\frac{e^{4t} + 1}{2} \right) - 5t$$

depending on whether
you take out e^{2t} or e^{-2t}

as a factor!

Question (7)

(b)



(i) For a circular arrangement

$$\frac{8!}{8} = 7!$$

(ii) Fix one man as a ref. (one way) The rest of the men can arrange themselves in $4!$ way.

The women can arrange themselves between the men in $(5 \times 4 \times 3)$ ways.

$\therefore$ The number of ways to arrange 5M, 3W so that no two women are next to one another

$$= \frac{(4!) \times (5 \times 4 \times 3)}{5! \times 12} = 1440$$

(c) $x^3 - mx + n = 0$ — (1)

(i) Let $y = x^3 \Rightarrow x = y^{1/3}$

$\therefore$ (1) becomes $y - my^{1/3} + n = 0$

$$\Rightarrow y^{1/3} = \left(\frac{y+n}{m} \right)$$

$$\therefore m^3 y = (y+n)^3$$

$$m^3 y = y^3 + 3ny^2 + 3n^2 y + n^3$$

Collecting like terms

$$y^3 + 3ny^2 + (3n^2 - m^3)y + n^3 = 0$$

(ii)

$$\alpha^6 + \beta^6 + \gamma^6 = (\sum \alpha_i)^2 - 2 \sum_{i \neq j} \alpha_i \alpha_j$$

$$\begin{aligned} (\sum \alpha_i)^2 &= (-3n)^2 - 2(3n^2 - m^3) \\ &= 9n^2 - 6n^2 + 2m^3 \end{aligned}$$

$$\alpha\beta\gamma = -n, \text{ (from } x^3 - mx + n = 0)$$

$$\therefore \alpha\beta\gamma(\alpha^6 + \beta^6 + \gamma^6) = -n(3n^2 + 2m^3)$$

Question 4)

$$(a) \quad 1 + p + p^2 + p^3 + p^4 = \frac{p^5 - 1}{p - 1} = 0 \quad \text{--- (1)}$$

$$p^5 = \left(\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} \right)^5$$

$$(i) \quad = \cos 2\pi + i \sin 2\pi \quad \left(\begin{array}{l} \text{De Moivre's} \\ \text{Thm} \end{array} \right)$$

$$= 1$$

$$\therefore \frac{p^5 - 1}{p - 1} = \frac{1 - 1}{p - 1} = 0$$

$$(ii) \quad (p + p^4)(p^2 + p^3)$$

$$= p^3 + p^4 + p^6 + p^7 \quad [\because p^5 = 1]$$

$$= p^3 + p^4 + p + p^2$$

$$\text{from (1) } \boxed{1 + p + p^2 + p^3 + p^4 = 0}$$

$$= -1 \quad \text{--- (2)}$$

(iii) The quadratic equation is:

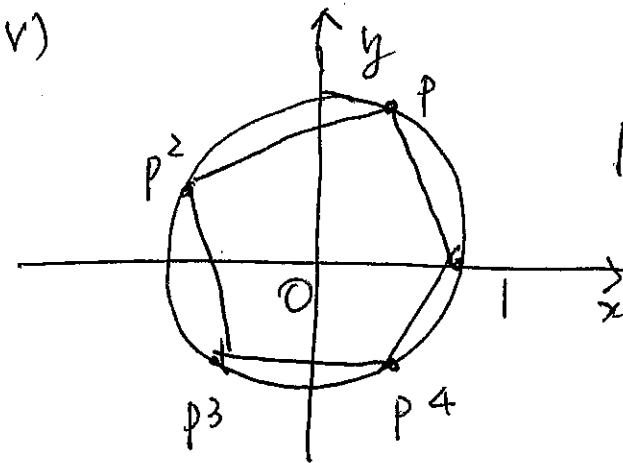
$$z^2 - [(p + p^4) + (p^2 + p^3)]z + (p + p^4)(p^2 + p^3) = 0$$

From (1) & (2)

$$z^2 - (-1)z + (-1) = 0$$

$$\text{i.e. } \boxed{z^2 + z - 1 = 0}$$

(iv)



$$p^4 = \left(\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} \right)^4$$

$$= \cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5}$$

$$= \cos \left(-\frac{2\pi}{5} \right) + i \sin \left(-\frac{2\pi}{5} \right)$$

$$= \bar{p}$$

$$\text{Also, } p^3 = \left(\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} \right)^3$$

$$= \cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5} = \cos \left(-\frac{4\pi}{5} \right) + i \sin \left(-\frac{4\pi}{5} \right) = \bar{p}^2$$

$$\text{Now, } (p + p^4)(p^2 + p^3) = -1 \quad \left(\begin{array}{l} \text{proven} \\ \text{from (ii)} \end{array} \right)$$

$$\therefore (p + \bar{p})(p^2 + \bar{p}^2) = -1$$

$$-1 = 2 \operatorname{Re}(p) \times 2 \operatorname{Re}(p^2)$$

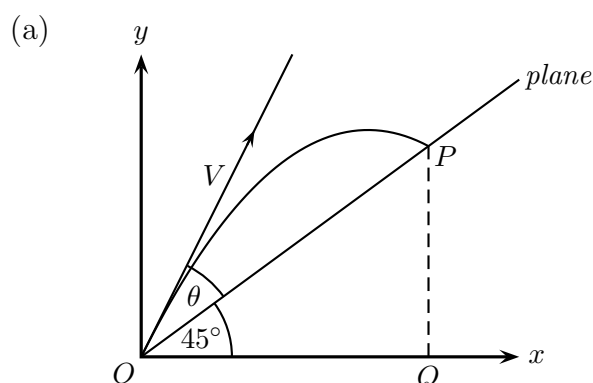
$$-1 = 4 \cos \frac{2\pi}{5} \cos \frac{4\pi}{5}$$

$$\text{i.e. } \boxed{\cos \frac{2\pi}{5} \cos \frac{4\pi}{5} = -\frac{1}{4}}$$

2014 Extension 2 Mathematics Task 2:
Solutions— Section C

Marks

Question 5 (14 marks)



A particle is projected from a point O on a plane inclined at 45° to the horizontal. The velocity of projection is V at an angle of θ to the inclined plane, where $\theta = \tan^{-1} \frac{1}{2}$.

[You are given that $y = x \tan \alpha - \frac{gx^2}{2V^2}(1 + \tan^2 \alpha)$ where $\alpha = 45^\circ + \theta$.]

(i) Show that $\tan \alpha = 3$.

1

Solution: $\tan \alpha = \tan(\theta + 45^\circ),$

$$= \frac{\frac{1}{2} + 1}{1 - \frac{1}{2} \times 1},$$

$$= 3.$$

(ii) Hence show that $y = 3x - \frac{5gx^2}{V^2}$.

1

Solution: $y = 3x - \frac{gx^2}{2V^2}(1 + 9),$

$$= 3x - \frac{10gx^2}{2V^2},$$

$$= 3x - \frac{5gx^2}{V^2}.$$

(iii) Show that the range OP is $\frac{2\sqrt{2}V^2}{5g}$.

2

Solution: At P , $y = x$,
 so $2x = \frac{5gx^2}{V^2},$

$$x = \frac{2V^2}{5g} \text{ as } x \neq 0.$$

 Now $OP = \sqrt{x^2 + x^2},$

$$= \sqrt{2}x,$$

$$= \frac{2\sqrt{2}V^2}{5g}.$$

(iv) Show that it meets the plane at right-angles.

Solution: Method 1—

$$\frac{dy}{dx} = 3 - \frac{10gx}{V^2}.$$

$$\text{At } P, x = \frac{2V^2}{5g},$$

$$\begin{aligned} \text{so } \frac{dy}{dx} &= 3 - \frac{10g}{V^2} \times \frac{2V^2}{5g}, \\ &= -1 \text{ at } P. \end{aligned}$$

Now the slope of the plane is 1, so the particle meets it at right-angles.

Solution: Method 2—

$$\begin{aligned} \ddot{y} &= -g & \ddot{x} &= 0 \\ \dot{y} &= -gt + V \sin \alpha & \dot{x} &= V \cos \alpha \\ y &= -\frac{gt^2}{2} + Vt \sin \alpha & x &= Vt \cos \alpha \end{aligned}$$

$$\text{At } P, Vt \cos \alpha = Vt \sin \alpha - \frac{gt^2}{2}, t \neq 0$$

$$\frac{gt}{2} = V \sin \alpha - V \cos \alpha$$

$$t = \frac{2V}{g} \times \left(\frac{3-1}{\sqrt{10}} \right)$$

$$= \frac{4V}{g\sqrt{10}}$$

$$\dot{y} = \frac{3V}{\sqrt{10}} - \frac{4V}{\sqrt{10}}$$

$$= -\frac{V}{\sqrt{10}}$$

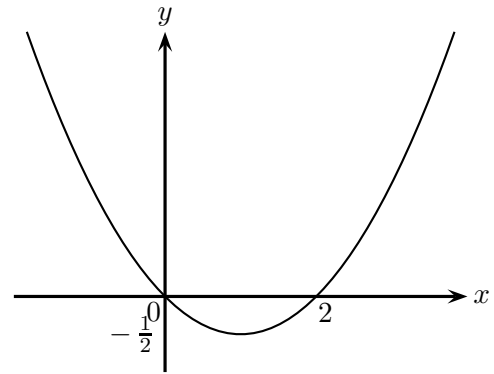
$$\begin{aligned} \therefore \frac{\dot{y}}{\dot{x}} &= -\frac{V}{\sqrt{10}} \times \frac{\sqrt{10}}{V} \\ &= -1. \end{aligned}$$

Now the slope of the plane is 1, so the particle meets it at right-angles.

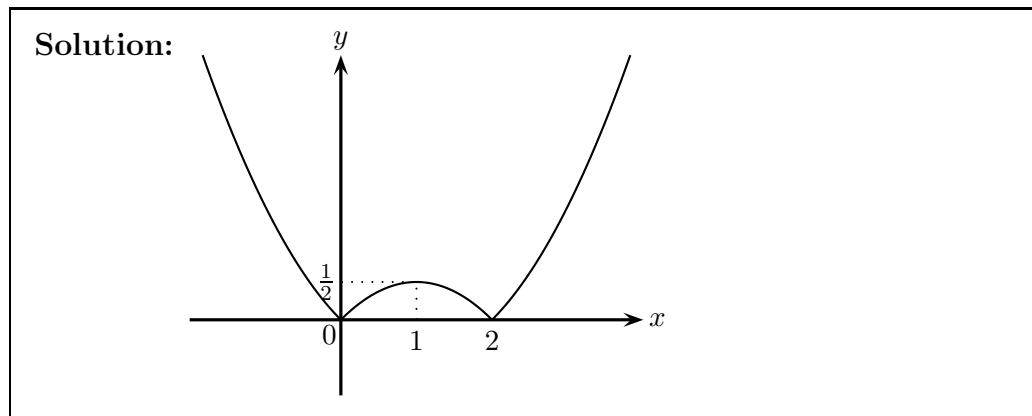
(b) The sketch is of the parabola

$$f(x) = \frac{x^2 - 2x}{2}.$$

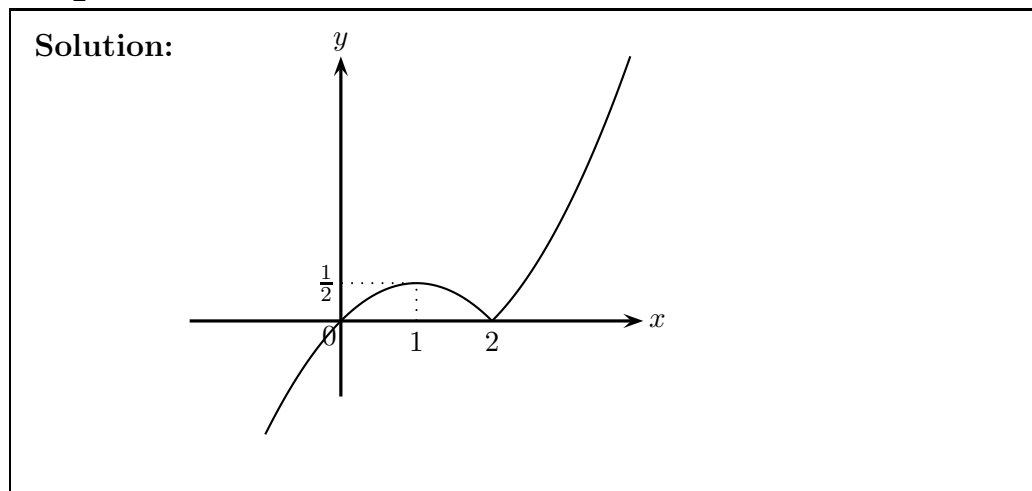
Without calculus, draw the following on separate diagrams, showing essential features:



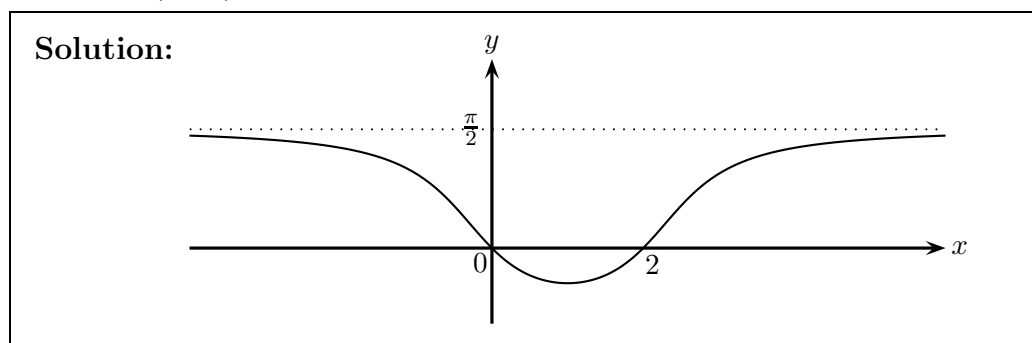
(i) $y = |f(x)|$



(ii) $y = \frac{x}{2}|x - 2|$

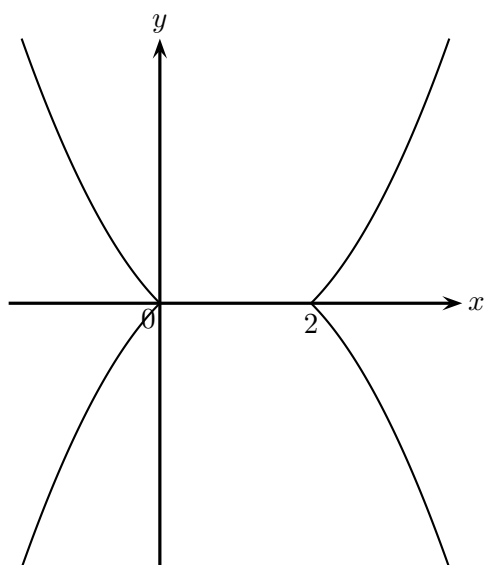


(iii) $y = \tan^{-1}(f(x))$



(iv) $|y| = f(x)$

Solution:

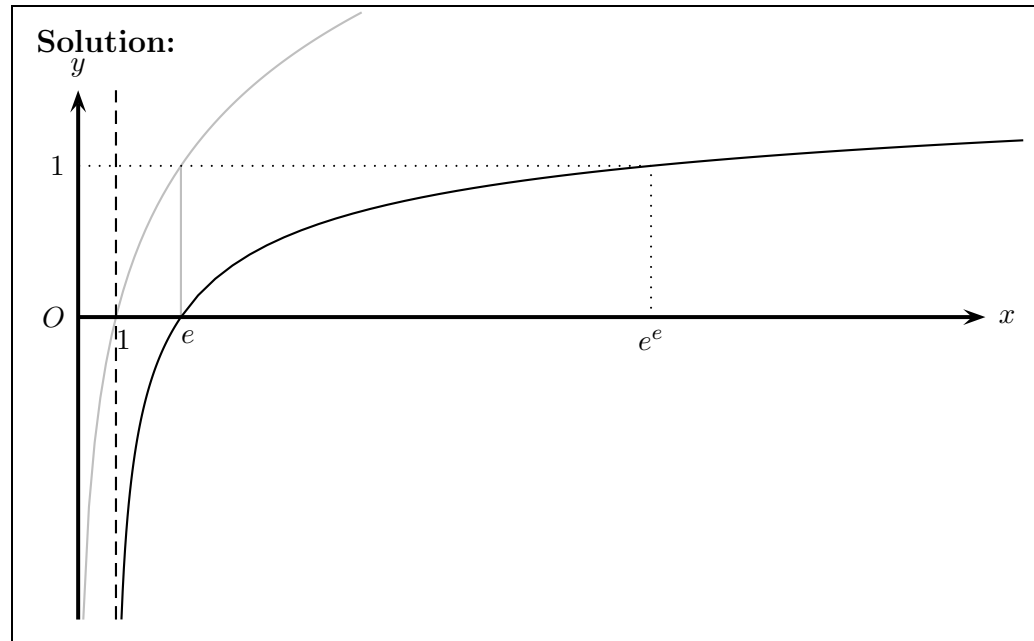


Question 6 (15 marks)

(a) Sketch the following, showing essential features:

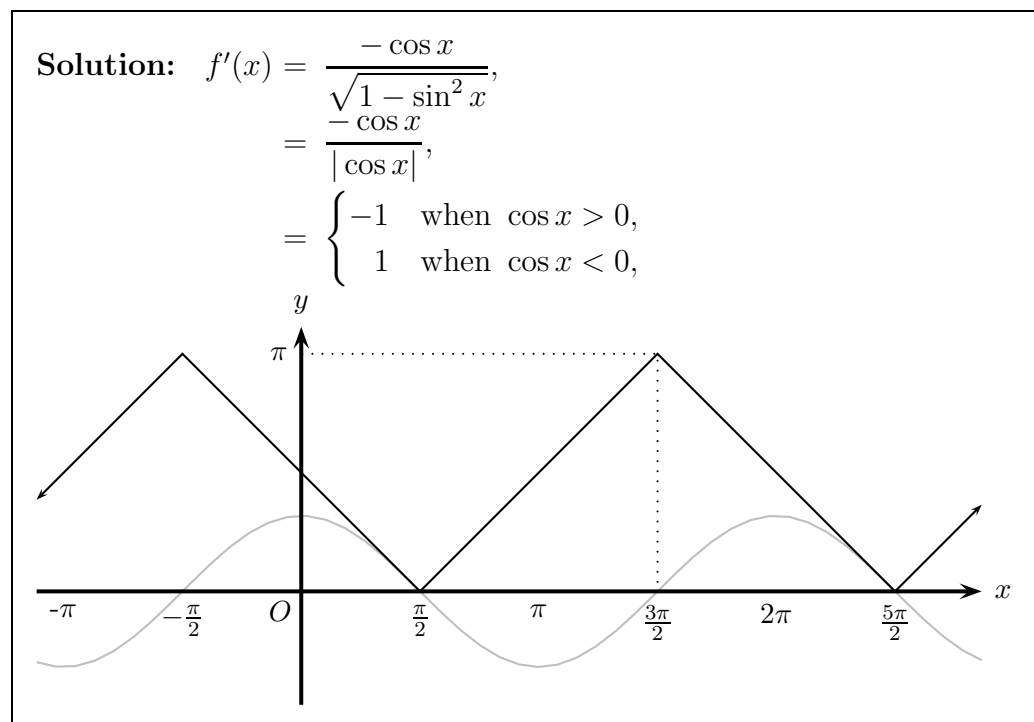
(i) $y = \ln(\ln x)$

2



(ii) $f(x) = \cos^{-1}(\sin x)$

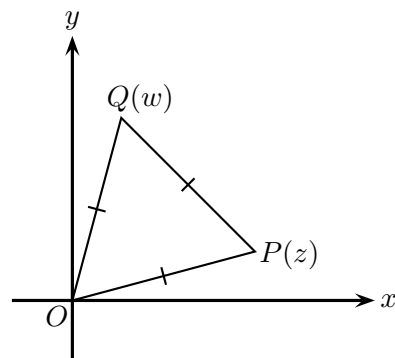
2



- (b) The triangle OPQ is equilateral.

P represents the complex number z , and
 Q represents the complex number w .

Show that $z^3 + w^3 = 0$.



Solution:

$$\begin{aligned} w &= z \operatorname{cis} \frac{\pi}{3}, \\ w^3 &= z^3 \operatorname{cis} \pi, \\ &= -z^3. \\ \therefore z^3 + w^3 &= z^3 - z^3, \\ &= 0. \end{aligned}$$

- (c) Let $I_n = \int_0^1 (1 - x^2)^n dx$ and $J_n = \int_0^1 x^2 (1 - x^2)^n dx$.

- (i) Using integration by parts on I_n , show that $I_n = 2nJ_{n-1}$.

Solution:

$$\begin{aligned} u &= (1 - x^2)^n & v' &= dx \\ u' &= -2xn(1 - x^2)^{n-1} & v &= x \end{aligned}$$

$$\begin{aligned} I_n &= \left[x(1 - x^2)^n \right]_0^1 + \int_0^1 2nx^2(1 - x^2)^{n-1} dx, \\ &= 0 + 2n \int_0^1 x^2(1 - x^2)^{n-1} dx, \\ &= 2nJ_{n-1}. \end{aligned}$$

- (ii) Hence show that $I_n = \frac{2n}{2n+1} I_{n-1}$.

Solution:

$$\begin{aligned} I_n &= -2n \int_0^1 -x^2(1 - x^2)^{n-1} dx, \\ &= -2n \int_0^1 \left((1 - x^2)(1 - x^2)^{n-1} - (1 - x^2)^{n-1} \right) dx, \\ &= -2n \int_0^1 (1 - x^2)^n dx + 2n \int_0^1 (1 - x^2)^{n-1} dx, \\ &= 2nI_{n-1} - 2nI_n, \\ (1 + 2n)I_n &= 2nI_{n-1}, \\ \therefore I_n &= \frac{2n}{2n+1} I_{n-1}. \end{aligned}$$

(iii) Show that $J_n = I_n - I_{n+1}$.

2

Solution: Method 1—

$$\begin{aligned} I_n - I_{n+1} &= \int_0^1 (1-x^2)^n dx - \int_0^1 (1-x^2)^{n+1} dx, \\ &= \int_0^1 \left((1-x^2)^n - (1-x^2)(1-x^2)^n \right) dx, \\ &= \int_0^1 x^2 (1-x^2)^n dx, \\ &= J_n. \end{aligned}$$

Solution: Method 2—

$$\begin{aligned} 2nJ_{n-1} &= \frac{2n}{2n+1} I_{n-1} \text{ (from (i) and (ii))}, \\ \therefore 2nJ_{n-1} + J_{n-1} &= I_{n-1}, \\ I_n + J_{n-1} &= I_{n-1}, \\ J_{n-1} &= I_{n-1} - I_n, \\ \text{so } J_n &= I_n - I_{n+1}. \end{aligned}$$

(iv) Hence deduce that $J_n = \frac{1}{2n+3} I_n$.

2

Solution: Method 1—

$$\begin{aligned} I_{n+1} &= (2(n+1))J_n \text{ (from (i))}, \\ \therefore J_n &= I_n - (2n+2)J_n \text{ (from (iii))}, \\ J_n(2n+2+1) &= I_n, \\ J_n &= \frac{1}{2n+3} I_n \end{aligned}$$

Solution: Method 2—

$$\begin{aligned} I_{n+1} &= \frac{2(n+1)}{2(n+1)+1} I_{(n+1)-1} \text{ (from (ii))}, \\ &= I_n \left(\frac{2n+2}{2n+3} \right). \\ \text{So } J_n &= I_n \left(\frac{2n+3 - (2n+2)}{2n+3} \right) \text{ (from (iii))}, \\ &= \frac{1}{2n+3} I_n. \end{aligned}$$

(v) Hence find a reduction formula for J_n in terms of J_{n-1} .

1

$$\begin{aligned} \textbf{Solution: } J_n &= \frac{1}{2n+3} \times 2nJ_{n-1}, \\ &= \frac{2n}{2n+3} J_{n-1}. \end{aligned}$$